Temperature-sensing riboceptors

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Abstract

Understanding how cells sense temperature is a fundamental question in biology and is pivotal for the evolution of life. In numerous organisms, heat is not only sensed but also generated due to cellular processes. Consequently, the mechanisms governing heat and cold sensation in various microorganisms and animals have been experimentally elucidated. But the use of the term thermoreceptor is largely limited to membranal proteins such as temperature-sensing TRP (transient receptor potential) channels. Yet, in addition to these membrane receptors, temperature sensing must occur in every organelle in the cell. Utilizing a colonial India 'punkah-wallahs' analogy, I present my rationale for the necessity of heat sensing in every organelle in a cell. Finally, I propose temperature-sensing **riboceptors** (**ribo**nucleic acid re**ceptors**) to integrate all the RNA molecules (mRNA, non-coding RNA and so forth) capable of sensing temperature and triggering a signaling event. This approach could enable the identification of riboceptors in every cell of almost every organism, not only for temperature but also for other classes of ligands.

Temperature is one of the pivotal regulators of evolution of molecular machinery and life. The 600 million years of evolutionary conservation of temperature regulation capacity have been suggested to provide a survival benefit in organisms.[1] Temperature can affect growth rates across various cells and organisms.[2-6] Apart from environmental changes in temperature, pathogen infections and toxin exposure in plants and animals can also increase organismal temperatures.[1,7-10] The mechanisms that allow organisms to survive high temperatures has been reported to be due to heat shock proteins, temperature-sensitive gene cluster architecture, differential osmotic activity, differential antioxidant defense mechanisms, differential nucleotide sequences that allow the formation of a relatively higher number of protein disulfide bonds, and protein stabilization.[11-13]

The mechanisms that allow cells to sense external temperature have been well studied. Temperature-sensing TRP channels (transient receptor potential) are well known as sensory thermoreceptors.[14-16] Temperature-sensing TRP channel proteins have relatively high Q_{10} (temperature coefficient: the fold-increase in rate per 10°C increase) in comparison to other non-temperature sensing channel proteins.[17] Similar to temperature-sensing mammalian TRP channels, *Drosophila melanogaster* dTrpA1 in the warmth-activated anterior cell neurons has been reported as 'molecular sensor of warmth'.[18] Another protein, GR28b.d in the antennal arista, was also reported as a 'thermosensor' in Drosphila.[19,20] In insect triatomine (*Rhodnius prolixus*), TRPA5B was reported to be a 'temperature-activated receptor' that exhibit a high temperature coefficient (Q_{10} = 25).[21]

But temperature differences exist not only in the external environment surrounding the organism but also internally across different regions and cellular organelles. The temperature of the nucleus is marginally higher than the cytoplasm.[22] In neuron-like cells *in vitro*, cell body temperature is higher than temperature in neurite-like structures.[23] Finally, in highly energy-active organelles such as mitochondria, the temperature is around 50℃ and ATP synthesis perturbations in mitochondria can also cause temperature differences.[24-26]. But why then the term thermoreceptors are largely used only for TRP channels in the plasma membrane?

As an analogy, during the colonization of India by the British, the British found it challenging to get acclimatized to the Indian heat while still wearing their aristocratic robes. To solve this issue, they took advantage of 'punkah wallahs' (ceiling fan coolies, servants, or slaves) who will rhythmically pull the punkahs via a pulley system. Punkahs are rectangular wooden frames with cloth that are suspended from the ceiling. As the British 'aristocrats' moved

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between their rooms, other punkah wallahs came into action, or punkah wallahs also had to move along to pull other punkahs linked to that specific room.[27] In some cases, the punkah wallahs were completely isolated from the room, as a small hole in the room allowed the rope to be pulled via the pulley. Even after the invention of electricity and air conditioners, air conditioners still require temperature sensors to sense the temperature in the room that needs to be cooled. Likewise, intrinsic heat-sensing mechanisms must also exist in almost every organelle. But then, what are the temperature-sensing receptors in the intracellular environment? We do have several tools to measure organelle temperature, but what is the identity and function of temperature-sensing receptors in different organelles?[28,29]

The catalytic activity of all enzymes decreases at increased temperatures (16-fold per 25℃).[30] Nevertheless, the outlier proteins with extreme differences in temperature-dependent activity are likely candidates for either heat-sensing or cold-sensing proteins. For instance, bacterial sigma 32 and DesK are intrinsic temperature-sensing protein with transcriptional and kinase activity, respectively.[31,32] In Salmonella, TlpA temperature sensing activity depends on the temperature-dependent monomer-to-coiled-coil equilibrium.[33] Another example is the TdcA (thermosensory diguanylate cyclase), a temperature-sensing protein that exhibits a 100-fold increase in the activity of c-di-GMP generation upon a 10℃ increase.[34] TdcA diguanylate cyclase is inactive at 22℃ and highly active at 37℃. TdcA and some TdcA homologs are thermosensitive via a conserved temperature-sensitive PAS (Per-Arnt-SIM) domain. Human HSF1 (heat shock factor 1) was also identified as an intrinsic temperature-sensing protein.[35] HSF1 is a transcription factor that trimerizes at increased temperature, binds to promoter sites with heat shock elements, and induces gene expression.

At present, there is inconsistency in the terminology used to refer to temperature-sensing signaling proteins. Terms found in temperature-sensing literature include 'thermosensor,' 'thermometer', and 'thermal receptor'. However, the terms 'temperature-sensing receptor' or 'thermoreceptor' are primarily used when referring to temperature-sensing TRP channels. It is unclear why sigma 32, TlpA, TdcA, and HSF1 are not referred to as temperature-sensing receptors or thermoreceptors. This lack of unity reminds me of the blind men and the elephant analogy that I also stressed while proposing the need for the term gasoreceptors for gas-sensing proteins.[36-38] If we need to differentiate between 'sensory thermoreceptors' and internal 'temperature-sensing receptors,' instead of using term 'non-sensory thermoreceptor', I propose the term 'agnireceptors' ('Agni' in Sanskrit

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for 'God of fire') for all proteins, regardless of their localization, that can sense temperature and trigger a signal.

Moreover, temperature-sensing must be essential in organisms that lack proteins as well. To understand this, we can learn from RNA-based temperature-sensing mechanisms previously reported in microorganisms. For instance, some mRNAs have temperature-sensing elements that allow translation of downstream genes by blocking ribosome-binding sites.[39-42] An example is the temperature-sensitive 5' mRNA coding sequence of the bacterial rpoH gene, which codes for the heat shock transcription factor sigma 32.[43] Another example is the lcrF Shine-Dalgarno sequence in the *Yersinia pestis* [44]. Similarly, temperature-sensitive ncRNA (non-coding RNA) *COOLAIR, COLDWRAP* and *COLDAIR* in plants have also been reported.[45,46] All these are referred to as 'RNA thermometers' or 'RNA thermosensors'. In my opinion, they are merely RNA-based temperature-sensing receptors or proto-receptors. We must debate about when should we call it a receptor, and when should we not? For instance, riboswitches have also been proposed as metabolite and hormone receptors, so why not for temperature as well?[47,48] How many downstream signaling events must occur to call an RNA thermometer or thermosensor as an RNA-based temperature-sensing receptor or thermoreceptor? We have the term 'ribozymes' for RNA-based enzymes, so why not have a unifying term for RNA-based receptors?[49]

To encompass all RNA molecules (mRNA, non-coding RNA and so forth) capable of directly sensing various stimuli within a cell (temperature, gravity, gas, water, metal ions, amino acids, pH, other biological molecules, sound and so forth) and triggering a signal, I propose the term 'riboceptors' (**ribo**nucleic acid re**ceptors**). Examples of riboceptors include RNA-based evolutionarily old proto-receptors, such as the lithium-, sodium-, thiamine-, lysine-sensing riboswitches.[50-53] For temperature-sensing receptors, temperature *per se* is the agonist and not an allosteric modulator. Finally, identifying intrinsic temperature-sensing protein-based agnireceptors and RNA-based riboceptors may help us better understand processes such as RNA editing and/or the cross-talk with H_2S-sensing gasoreceptors in gasocrine signaling and animal physiology.[34,37,54,55]

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CONFLICT OF INTEREST

None.

DISCLOSURE

The author used ChatGPT for correcting the scientific English. The author takes full responsibility for the content of this manuscript.

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